

ANTHONY BWEMBYA

SENSOR SYSTEMS | SCIENTIFIC COMPUTING | AI & DATA

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PROFESSIONAL PROFILE

Experimental physicist and sensor-data specialist with experience developing, calibrating and validating complex radio, optical and detector-based measurement systems. Strong background in Python/C++ scientific software, signal reconstruction, sensor-response modelling, uncertainty analysis, automated quality control and machine learning for physical measurements. Combines hands-on instrumentation with reproducible, data-driven analysis.

CORE COMPETENCIES

Scientific programming & data

Python, C++, Bash, SQL, NumPy, SciPy, pandas, Matplotlib, Jupyter, Linux, Git, HPC/SLURM

Machine learning & AI

PyTorch, scikit-learn, feature engineering, classification, regression, clustering, anomaly detection and model validation

Signal processing

RF processing, Fourier and spectral analysis, phase calibration, timing synchronisation, coherent reconstruction and noise modelling

Sensors & instrumentation

Antenna arrays, detector calibration, LiDAR, fluorescence telescopes, PMTs, radiation detectors, DAQ and RF/DC chains

Modelling & quality

Statistical inference, uncertainty propagation, CORSIKA/CoREAS, GALPROP, model-to-data comparison and automated quality control

Collaboration & delivery

Technical documentation, scientific communication, stakeholder engagement, teaching, mentoring, safety and field operations

PROFESSIONAL EXPERIENCE

Sensor Development, Calibration and Intelligent Data Processing

Radboud University / Pierre Auger Observatory · Netherlands / Argentina · 2021–2026

- Developed precision calibration and synchronisation methods for a distributed detector array, achieving nanosecond-level timing alignment across autonomous stations.
- Built Python and C++ workflows for waveform processing, spectral analysis, feature extraction, reconstruction, sensor diagnostics and automated quality control.
- Designed network-level consistency tests to identify anomalous sensors, timing drift, geometry errors, ambiguous solutions and inconsistent measurements.
- Modelled wave propagation, detector geometry, atmospheric refractivity and antenna response to validate reconstructed events against physical expectations.
- Developed coherent interferometric reconstruction methods for weak signals, culminating in the first successful reconstruction of a highly inclined event with AugerPrime radio-detector data.
- Supported LiDAR and fluorescence-detector calibration and operations, including atmospheric monitoring and detector-performance assessment.

PROFESSIONAL EXPERIENCE CONTINUED

Research & Technical Officer - Radiation Measurements and Instrumentation

Radiation Protection Authority, Zambia · 2021–2022

- Performed radiation and spectroscopic measurements, instrument checks, detector calibration, sample preparation, spectral interpretation and technical reporting.
- Investigated anomalous readings, evaluated measurement uncertainty, and supported field inspections and compliance documentation.

Teaching Assistant - Machine Learning, Physics & Scientific Computing

Radboud University · 2021–2025

- Supported teaching in machine learning for particle physics and astrophysics, experimental physics, scientific programming and data interpretation.
- Guided students in model evaluation, feature-based analysis, uncertainty, Python workflows and technical communication.

Research Intern - Radiation Detection and Materials Analysis

National Institute for Scientific and Industrial Research, Zambia · 2017

- Performed alpha/gamma spectroscopy, X-ray fluorescence measurements, sample preparation and laboratory calibration.

EDUCATION

Doctoral Research in High-Energy Physics

Radboud University, Netherlands · 2021–2026

- Thesis: *In Phase With the Cosmos: Mass Composition of Cosmic Rays via Radio Interferometry*. Thesis completed and approved for defence.

MSc, Nuclear Physics and Technology

National Research Nuclear University MEPhI, Russia · 2018–2021

- Investigated the high-energy positron excess using PAMELA and AMS-02 observations with GALPROP propagation modelling.

BSc, Physics

University of Zambia · 2013–2017

- Broad foundation in mechanics, electromagnetism, quantum mechanics, statistical physics, electronics, solid-state physics, renewable energy and computational physics.

LANGUAGES

English Fluent - international working language	Bemba Native
Nyanja Native	Russian Professional working level
Dutch Developing	Work eligibility No visa sponsorship required in the Netherlands